

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 12
Matrices.

Review

1. What is the *orthonormal expansion* of $\vec{x} = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$ with respect to the orthonormal list $\vec{e}_1, \vec{e}_2, \vec{e}_3$?
2. Apply the Gram-Schmidt algorithm to the input set $\vec{a}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \vec{a}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.
3. Apply the Gram-Schmidt algorithm to the input set $\vec{a}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \vec{a}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \vec{a}_3 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$.
What do we learn?

New (Chp. 6)

1. Let M be the 3×2 matrix:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Write out $M_{i,j}$ for $i = 1, 2, 3$ and $j = 1, 2$.

2. Write M as a matrix of column vectors $\begin{bmatrix} \vec{a}_1 & \vec{a}_2 \end{bmatrix}$. What are $\vec{a}_1$ and $\vec{a}_2$?

3. Write M as a matrix of row vectors vectors $\begin{bmatrix} \vec{b}_1 \\ \vec{b}_2 \\ \vec{b}_3 \end{bmatrix}$. What are $\vec{b}_1, \vec{b}_2$ and $\vec{b}_3$?

4. What is the matrix $0_{3,2}$? Is it the same as $0_{2,3}$?

5. Find the diagonal matrix $\text{diag}(3, 4, 5)$.

6. Find a 3×3 *upper triangular* matrix U with all non-zero entries equal to 2.

7. Find a 3×3 *lower triangular* matrix L with all non-zero entries equal to 3.

8. Let M be the 3×2 matrix:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Find the *transpose* M^T .

9. Find the scalar product $5M$.

10. Let N be the 3×2 matrix:

$$\begin{bmatrix} -1 & 0 \\ 3 & 4 \\ 7 & 8 \end{bmatrix}$$

Find $M + N$.

11. Find $\|M\|$.